Exploratory Data Analysis:

Automobile Dataset

Report

Written by: Perryn Richardson

Introduction

The automobile dataset covers 205 cars across 26 variables, including make, price, engine size, horsepower, and fuel economy figures. The dataset was sourced as a text file and required cleaning before any meaningful analysis could be done.

Data Cleaning

Two columns were dropped upfront, ‘symboling’ and ‘normalized-losses’, as neither was relevant to the questions being investigated. The data was checked for duplicates; none were found.

Missing values were not stored as blanks. Instead, they appeared as '?' characters, which pandas read as valid text. These were replaced with ‘NaN’ before any rows with missing data were removed. Six columns were affected: num-of-doors, bore, stroke, horsepower, peak-rpm, and price. Dropping those rows brought the dataset down from 205 to 193.

Several columns that should have been numeric were stored as text, a side effect of the '?' issue. These were converted: whole number columns to int64 and decimal columns to float64.

Missing Data

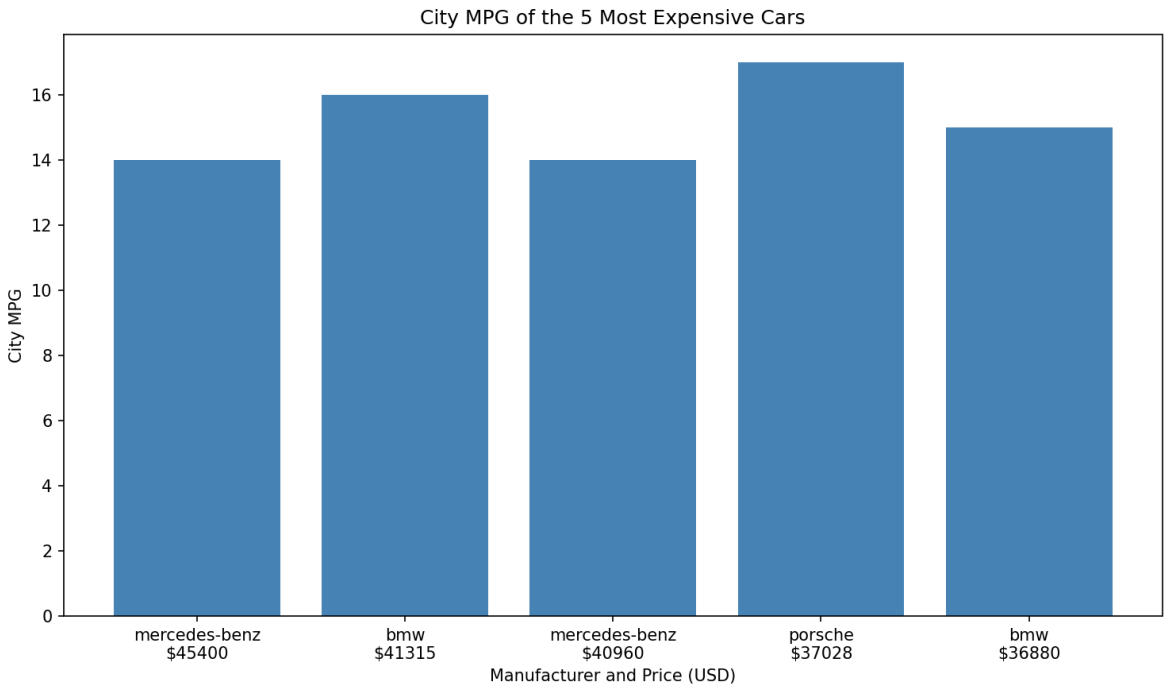
The missing data was spread across six columns, with between 2 and 4 missing values each. Rather than filling in estimated values for mechanical measurements like bore and stroke, the affected rows were removed. Guessing at engine specifications would have introduced errors into the analysis.

Data Stories and Visualisations

1) Which automobiles are in the hatchback category?

Filtering by body style returned 63 hatchback vehicles. Toyota, Volkswagen, and Dodge were the most represented manufacturers in this category. Hatchbacks made up the largest single body style group in the cleaned dataset.

2) Which are the 5 most expensive cars?



The five most expensive cars all come from three European manufacturers: Mercedes-Benz, BMW, and Porsche, with prices between $36,880 and $45,400. None of them return strong fuel economy figures, ranging from 14 to 17 city MPG. These cars are built for performance, not efficiency.

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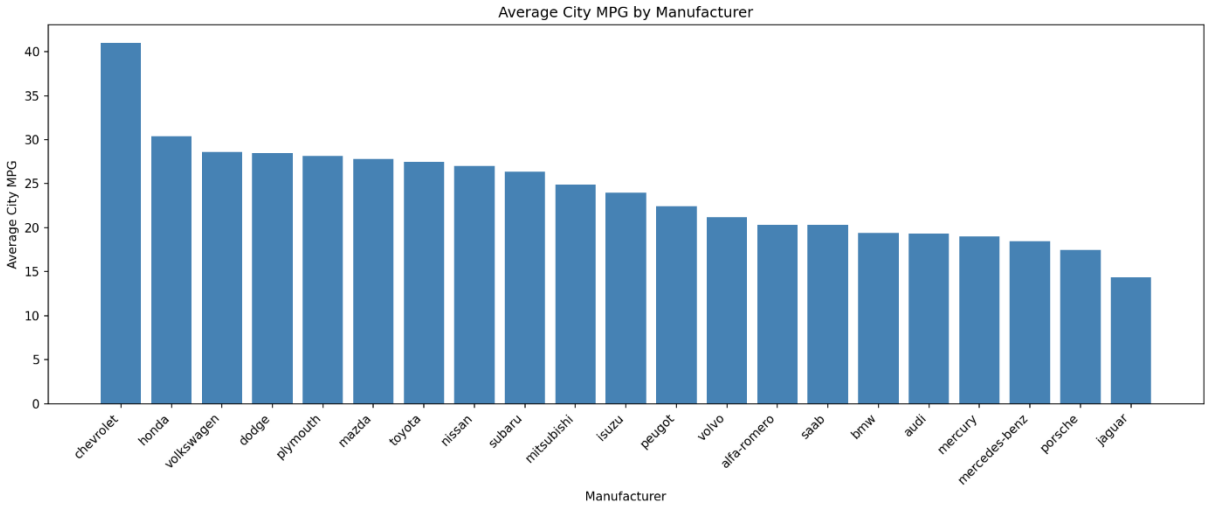
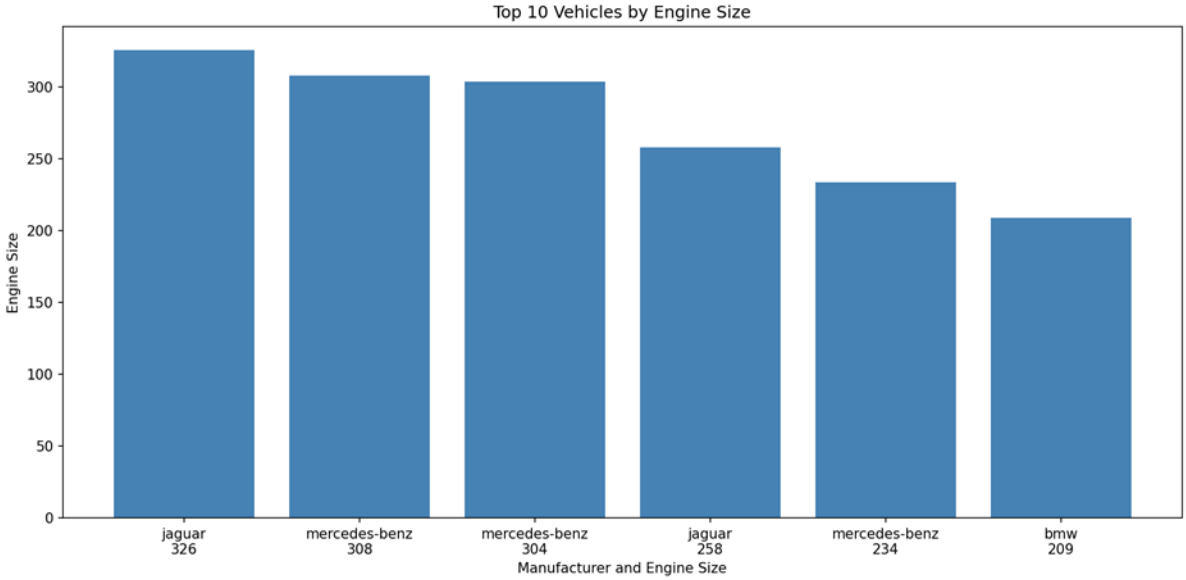
3) Which manufacturer builds the most fuel-efficient vehicles?



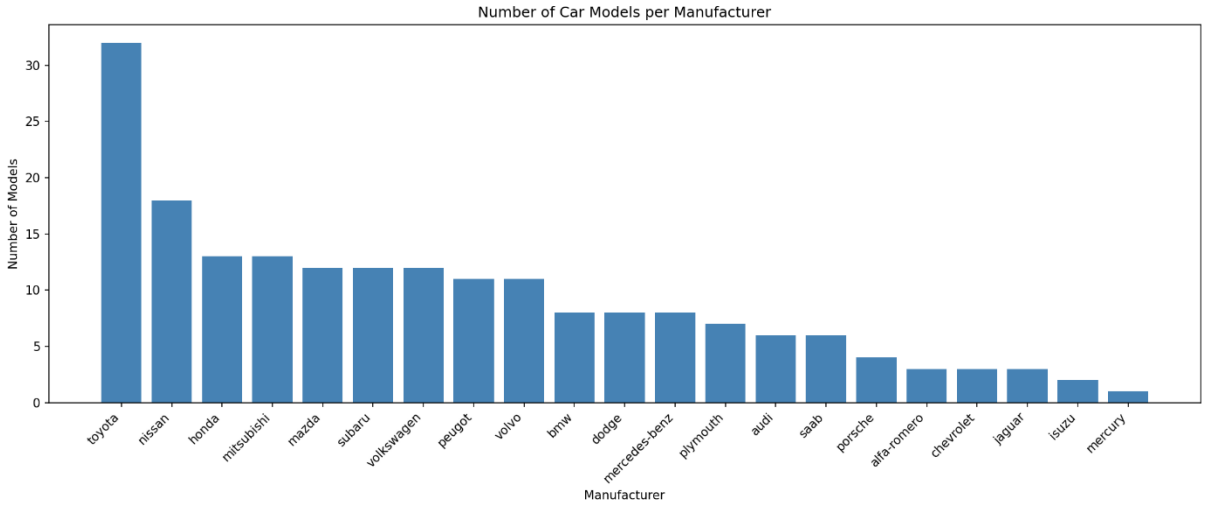
Chevrolet sits well clear at the top with an average of around 41 city MPG. Honda and Volkswagen follow. At the other end, Jaguar, Porsche, and Mercedes-Benz return the lowest figures. These are the same brands that appeared at the top of the price and engine size charts

4) Which vehicles have the largest engine capacity?



Jaguar has the largest engine in the dataset at 326 cubic inches. Mercedes-Benz takes the next two spots at 308 and 304. The top 10 is made up entirely of European luxury brands: Jaguar, Mercedes-Benz, and BMW. Bigger engines, higher prices, and lower fuel economy all point in the same direction.

5) Which vehicle manufacturer has the most car models in the dataset?



Toyota has 32 models in the dataset, more than any other manufacturer. Nissan comes second with 18. The gap between first and second is larger than the gap between second and tenth, reflecting how broadly Toyota was represented across different body styles and price points.

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